

Gauss Elimination Method:- (Direct method)

"Consider the system of n -linear Algebraic Equations with n unknowns $x_1, x_2, \dots, x_n$ "

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n = b_3$$

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = b_n$$

Write augmented matrix Ab

$$Ab = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} & b_n \end{bmatrix}$$

Reduce Ab into echelon form
(by using Row & Column operation).

$$Ab = \begin{bmatrix} 1 & t_{12} & t_{13} & \dots & t_{1n} & K_1 \\ 0 & 1 & t_{23} & \dots & t_{2n} & K_2 \\ 0 & 0 & 1 & \dots & t_{3n} & K_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & K_n \end{bmatrix}$$

"In This way we obtain the system"

$$x_1 + t_{12}x_2 + t_{13}x_3 + \dots + t_{1n}x_n = K_1$$

$$0 + x_2 + t_{23}x_3 + \dots + t_{2n}x_n = K_2$$

$$0 + 0 + x_3 + \dots + t_{3n}x_n = K_3$$

$$x_n = K_n$$

The remaining un-known are evaluated by Backward substitutions.

Example:- Solve the following system by Gauss elimination method-

$$2x_1 + 2x_2 + 4x_3 = 18$$

$$x_1 + 3x_2 + 2x_3 = 13$$

$$3x_1 + x_2 + 3x_3 = 14$$

Sol

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$$Ab = \begin{bmatrix} 2 & 2 & 4 & 18 \\ 1 & 3 & 2 & 13 \\ 3 & 1 & 3 & 14 \end{bmatrix}$$

Reduce Ab into echelon form

$$= \underline{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 1 & 3 & 2 & 13 \\ 3 & 1 & 3 & 14 \end{bmatrix} \begin{array}{l} \text{by} \\ \frac{1}{2} R_1 \end{array}$$

$$R_1 = \frac{1}{2} \cdot 2 \quad \frac{1}{2} \cdot 2 \quad \frac{1}{2} \cdot 4 \quad 18 \cdot \frac{1}{2}$$

$$\begin{array}{cccc} 1 & 1 & 2 & 9 \end{array}$$

$$= \underline{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & 0 & 4 \\ 0 & -2 & -3 & -13 \end{bmatrix} \begin{array}{l} \text{By} \\ R_2 - R_1 \\ R_3 + (-3)R_1 \end{array}$$

$$R_2 = \begin{array}{cccc} 1 & 3 & 2 & 13 \\ -R_1 = -1 & -1 & -2 & -9 \end{array}$$

$$R_2 - R_1 = \begin{array}{cccc} 0 & 2 & 0 & 4 \end{array}$$

$$R_3 = \begin{array}{cccc} 3 & 1 & 3 & 14 \\ -3R_1 = -3 & -3 & -6 & -27 \end{array}$$

$$R_3 - 3R_1 = \begin{array}{cccc} 0 & -2 & -9 & -13 \end{array}$$

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$$= \underline{R} \left[\begin{array}{cccc} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & -2 & -3 & -13 \end{array} \right] \begin{array}{l} \text{By} \\ \frac{1}{2} R_2 \end{array}$$

$$\frac{1}{2} R_2 = 0 \cdot \frac{1}{2} \quad 2 \cdot \frac{1}{2} \quad 0 \cdot \frac{1}{2} \quad 4 \cdot \frac{1}{2}$$

$$\frac{1}{2} R_2 = 0 \quad 1 \quad 0 \quad 2$$

$$= \underline{R} \left[\begin{array}{cccc} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -3 & -9 \end{array} \right] \begin{array}{l} \text{By} \\ R_3 + 2R_2 \end{array}$$

$$R_3 = 0 \quad -2 \quad -3 \quad -13$$

$$2R_2 = 0 \quad 2 \quad 0 \quad 4$$

$$R_3 + 2R_2 = 0 \quad 0 \quad -3 \quad -9$$

$$= \underline{R} \left[\begin{array}{cccc} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \begin{array}{l} \text{By} \\ -\frac{1}{3} R_3 \end{array}$$

$$\Rightarrow x_1 + x_2 + 2x_3 = 9 \quad \text{--- (i)}$$

$$x_2 = 2 \quad \text{--- (ii)}$$

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$$\Rightarrow x_3 = 3 \quad \text{--- (1)}$$

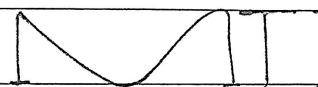
Put the value of x_2 & x_3 in Eq (1)

$$x_1 + 2 + 8 = 9$$

$$x_1 = 9 - 8 \Rightarrow x_1 = 1$$

Thus,

$$x_1 = 1, x_2 = 2, x_3 = 3$$

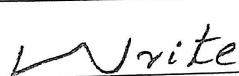
$\therefore$  : 603 :-

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Gauss JORDAN METHOD

Consider the system of Linear Equations

$$Ax = B$$

 Write the augmented matrix

Ab

Convert Ab into reduced Echelon form

$$\left[\begin{array}{cccccccc|c} 1 & 0 & - & - & - & - & - & 0 & K_1 \\ 0 & 1 & - & - & - & - & - & 0 & K_2 \\ 0 & 0 & 1 & - & - & - & - & 0 & K_3 \\ 0 & 0 & 0 & - & - & - & - & 1 & K_n \end{array} \right]$$

Which gives direct solution:-

$$x_1 = K_1, x_2 = K_2, x_3 = K_3, x_n = K_n$$

Example:- Solve the following system by Gauss Jordan method

$$2x_1 + 2x_2 + 4x_3 = 18$$

$$x_1 + 3x_2 + 2x_3 = 13$$

$$3x_1 + x_2 + 3x_3 = 14$$

Sol: Here

$$Ab = \begin{bmatrix} 2 & 2 & 4 & 18 \\ 1 & 3 & 2 & 13 \\ 3 & 1 & 3 & 14 \end{bmatrix}$$

$$= \underline{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 1 & 3 & 2 & 13 \\ 3 & 1 & 3 & 14 \end{bmatrix} \begin{array}{l} \text{By} \\ \frac{1}{2}R_1 \end{array}$$

$$= \underline{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & 0 & 4 \\ 0 & -2 & -3 & -13 \end{bmatrix} \begin{array}{l} R_2 - R_1 \\ R_3 - 3R_1 \end{array}$$

$$= \underline{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & -2 & -3 & -13 \end{bmatrix} \frac{1}{2}R_2$$

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$$= \underline{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -3 & -9 \end{bmatrix} \begin{array}{l} \text{By} \\ R_3 + 2R_2 \end{array}$$

$$= \underline{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{array}{l} \text{By} \\ -\frac{1}{3} R_3 \end{array}$$

$$= \underline{R} \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{array}{l} \text{By} \\ R_1 - 2R_3 \end{array}$$

$$= \underline{R} \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{array}{l} \text{By} \\ R_1 - R_2 \\ R_1 - R_3 \end{array}$$

$$\Rightarrow x_1 = -2, x_2 = 2, x_3 = 3$$